REVISIONS			
REVISION	**DESCRIPTION**	**DATE**	**MADE BY**
01	INITIAL RELEASE	2026/10/11	G. BARBIERI

SHEET 2

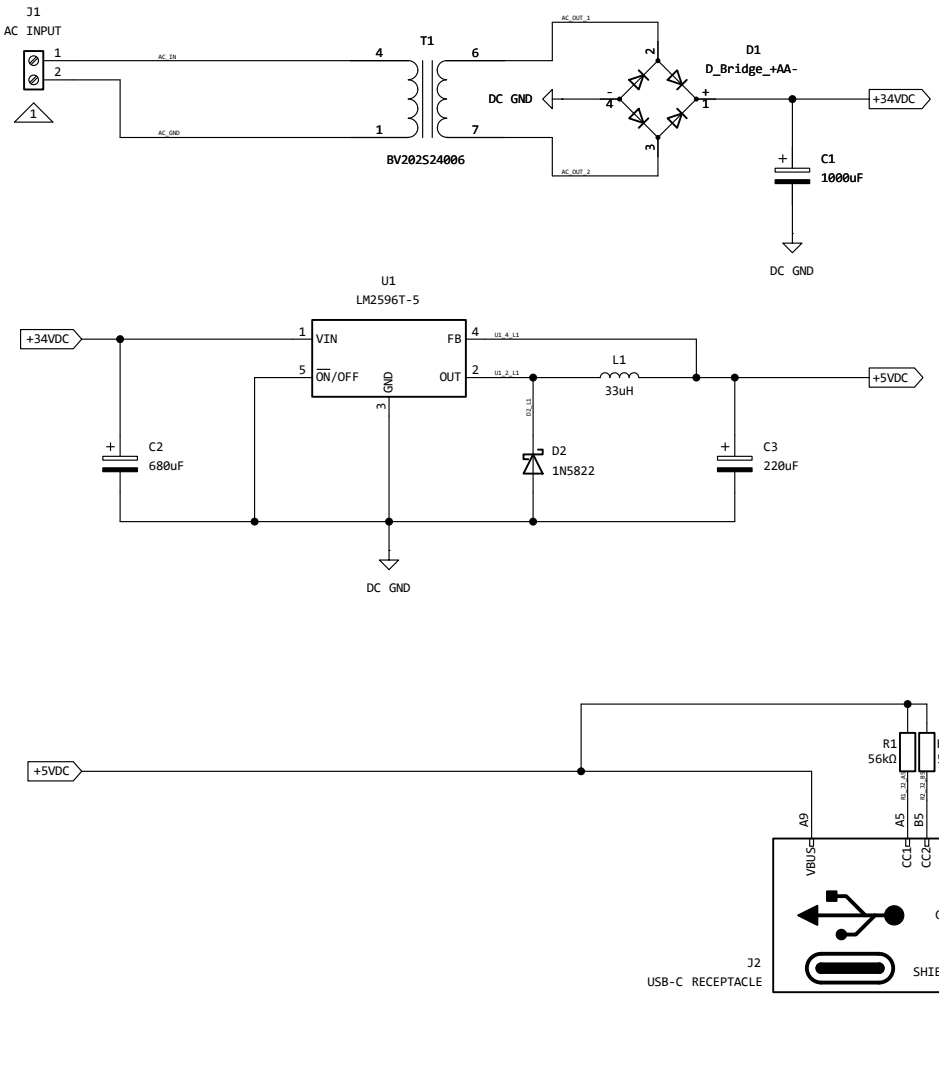
CHARGER SCHEMATIC

SHEET 3

PARTS LIST

TITLE: 5VDC/500mW - USB_C CHARGER SCH.
MADE BY: GUILHERME BARBIERI DA CUNHA
DATE: 2026-10-11
REV: 01
SIZE: A4
SHEET: 1/3
DRAWING NUMBER: ESCH-0001
BARBIERI INDUSTRIES PROPRIETARY

Barbieri Industries



1 AC INPUT IS 220VAC/60Hz

TITLE: 5VDC/500mW - USB_C CHARGER SCH.	
MADE BY: GUILHERME BARBIERI DA CUNHA	
DATE: 2026-10-11	
REV: 01	
SIZE: A4	
SHEET: 2/3	
DRAWING NUMBER: ESCH-0001	
BARBIERI INDUSTRIES PROPRIETARY	

Barbieri Industries

REFERENCE DESIGNATOR	DESCRIPTION	PART NUMBER	MANUFACTURER
C1	CAPACITOR (THT, 1000uF, 50V, 20%)	SL108M050HKJRR	NEXTGEN COMPONENTS
C2	CAPACITOR (THT, 680uF, 50V, 20%)	SL687M050HJJRR	NEXTGEN COMPONENTS
C3	CAPACITOR (THT, 220uF, 10V, 20%)	227CKS010M	CORNELL DUBILIER
D1	BRIDGE RECT. (4-DIP, 50V, 1A)	DF005M	DIODES INC.
D2	DIODE (THT, SCHOTTKY, 40V, 3A)	1N5822	STMICROELECTRONICS
J1	T.BLOCK (THT, SCREW)	TB007-508-02BE	SAME SKY
J2	USB-C RECEPTACLE (SMD, 16P)	TYPE-C-31-M-17	KOREAN HROPARTS ELEC
L1	INDUCTOR (THT, 33uH, 1,7A)	RLB0914-330KL	BOURNS INC.
R1, R2	RESISTOR (THT, 56kΩ, 5%, 1/2W, 1210)	CFM12JT56K0	STACKPOLE ELECTRONICS INC.
T1	TRANSFORMER (THT,22VAC/24VAC, 0.6VA)	BV202S24006	ZETTLER MAGNETICS
U1	IC CHIP (THT, 5V, 3A)	LM2596T-5.0/NOPB	TEXAS INSTRUMENTS

TITLE: 5VDC/500mW - USB_C CHARGER SCH.
MADE BY: GUILHERME BARBIERI DA CUNHA
DATE: 2026-10-11
REV: 01
SIZE: A4
SHEET: 3/3
DRAWING NUMBER: ESCH-0001
BARBIERI INDUSTRIES PROPRIETARY

Barbieri Industries